Reg. No. | | | | | | | | | | |

[12CS752]

SEVENTH SEMESTER B.E EXAMINATION DECEMBER–2015

DATA MINING

Time:3 Hours **Maximum Marks:70**

Instruction: Answer any **FIVE** full Questions.

1. a. State two factors that invokes the need for datamining. **02**
   b. State two differences between knowledge base and database. **02**
   c. Explain the following data pre-processing steps: **10**
      i. Aggregation
      ii. Sampling
      iii. Dimensionality Reduction
      iv. Feature subset creation
      v. Binarization

2. a. What is data visualization? Where is mainly used? **02**
   b. Explain the following visualization techniques: **10**
      i. Two dimensional histogram
      ii. Box plots
      iii. Star co-ordnates
      iv. Chernoff faces
      v. Matrices

   c. Comment on the use of a box plot to explore a data set with four attributes age, weight, height and income. **02**

3. a. Explain with an example decision trees. **02**
   b. Define gain ratio in the context of attribute splitting process. **02**
   c. Explain the building process of rule-based classifier with an example. **10**

4. a. Classification is a supervised learning Justify. **02**
   b. Define artificial neutral network (ANN). Explain multilayer ANN and perception model of ANN. **12**

5. a. How Bayesian theorem is used for classification? **04**
   b. Explain the concept of Naïve Bayesian and Bayesian belief networks. **10**

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6.	a.	Why ensemble is called classifier combination method?	02
	b.	Explain the concept of Bagging and Boosting.	06
	c.	Explain the operations of slicing, dicing, and drill-down in data cube.	06
7.	a.	Define support and confidence in mathematical terms.	02
	b.	Compare and constraint classification and clustering.	04
	c.	Write an algorithm for agglomerative hierarchical clustering. Explain the concept.	08
8.	a.	Explain the application of PCA in dimensionality reduction.	07
	b.	What is Regression? Explain simple Linear Regression with Least Square method.	07